

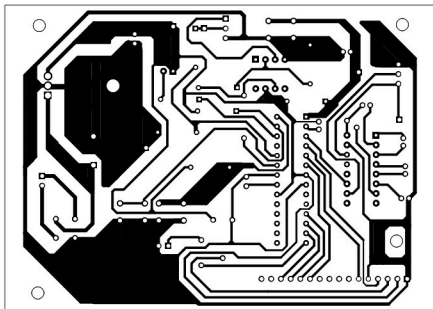


Fig. 9: An actual-size, single-side PCB for display circuit

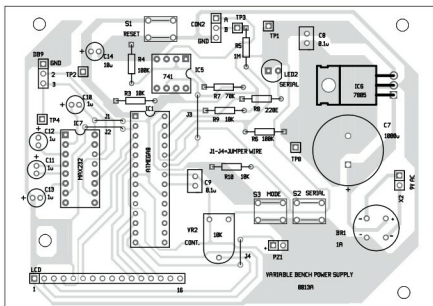


Fig. 10: Component layout for the PCB in Fig. 9

Visual Basic application.

User interface. The computer application is designed using Visual Basic 6.0—a rapid application development platform to develop GUI-based applications for Windows operating system. The screenshot of the application is shown in Fig. 6.

The application boosts an onboard digital display that reads the actual current, voltage and power readings from the hardware. It uses the MSCOMM32.ocx object to display the data on a

graphical chart. The chart is internally in sync with the latest reception of coded, framed string by the hardware. The string is decoded in the application and the current, voltage and power readings extracted. The application also displays the maximum value of current, voltage and power registered on the output terminal of the hardware.

To start the application, simply select the computer's COM port number from the dropdown menu to which your hardware is connected, then press

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

'Open' COM port button. The COM port is ready to receive data now.

On the hardware, press S2 momentarily. This will enable serial data transmission from the hardware. Now you will see that updation of reading has started in the application. You can stop it using the same button again. To reset the graph, press 'Reset' button. You can also change the strip chart view by using additional buttons provided on the left side of the application, such as WRAP/PAN mode, enable handles and enable grids. To stop the COM port, simply click 'Close' button.

If you want digital display fonts in the application, simply copy 'digifaw' font from the source code folder to the font folder in your computer and then restart the application.

Construction and testing

After you have completed assembly of the components on the PCBs, just plug these into the mains power socket and switch them on. You will hear two beeps and see the display loading. There are two modes of display, which can be changed by pressing S3 on the hardware. The first mode displays current and voltage only. Pressing switch S3 again will change the display to the second mode, where it displays the power. Switch S2 can be used to toggle serial data transmission 'on'/'off'.

Precautions. While using and designing the power supplies, the regulators heat up in case excess current is drawn from them. This situation can be handled by using proper-size and good-quality heat-sinks for the regulators. A fan may be fitted near the heat-sinks to cool them. No heat-sink is necessary for 7805 regulator (IC6) of the display section because the micro-controller circuit consumes less than 300mA current. ●

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